Child Vocalisation Context Classification

Group 05 Course: Advanced Machine Learning

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All code, results, and figures are available in the project repository.

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Abstract

We investigate the automated classification of child vocalisation contexts from short audio segments recorded across three languages (Shipibo-Conibo, Tagalog, Vietnamese). The task is formulated as 4-class supervised classification into PLTOY, MOVE, INTER, and EXTEV categories. We compare three models: a logistic regression (LR) baseline on Mel-frequency cepstral coefficient (MFCC) summary statistics, an audio-only Transformer with mean pooling (Model 2), and a Transformer with metadata-query attentive pooling (Model 3). All models are evaluated under a participant-disjoint holdout protocol with macro-F1 as the primary metric.

The headline finding is that LR on mel spectrogram summary statistics achieves the best test macro-F1 of any model. Systematic ablations show this result is robust across three feature sources, both pooling strategies, and all hyperparameter configurations tested. Error analysis confirms that 40.9% of test clips fail all three models simultaneously, indicating a data-driven ceiling. We trace the primary confound to a language mismatch between the validation and test children and quantify speaker-specific variance via Leave-One-Child-Out cross-validation ($\sigma_{\text{speaker}} \approx 0.053$ vs. $\sigma_{\text{seed}} = 0.000$ for deterministic LR).

As an extension, we apply Bayesian hyperparameter optimisation (Optuna TPE) across all 9 (model, feature) combinations. A continuous line search over $C \in [10^{-6}, 10^{6}]$ identifies $C = 0.00492$ for LR mel — falling between the discrete 9-point grid values — improving test macro-F1 from 0.277 to $\mathbf{0.311}$, the best result of the entire project. Two further genuine gains emerge: LR wav2vec2 improves from 0.263 to 0.285, and Model 2 on wav2vec2 improves from 0.219 to 0.251 ($+0.032$ at $|\Delta|/\sigma = 1.6$). The remaining six combinations are within seed noise of canonical or degrade. The pattern correlates with the canonical val-test gap: combinations with a small gap admit further tuning, while combinations with larger validation–test mismatch are dominated by the language confound, and Optuna can worsen overfitting to the Shipibo validation child.

Introduction

Understanding the contexts in which young children vocalise is of significant interest to developmental linguists and child language researchers. Automated methods that can classify vocalisation contexts from audio would reduce the burden of manual annotation, particularly in under-resourced language settings where annotated data are scarce.

This project addresses: *can we classify the interactional context of a child vocalisation from its acoustic content and limited metadata (age, gender)?* We work with a dataset of 1,009 clips from 9 children spanning three typologically distinct languages and four context categories.

The methodological contributions are:

- Logistic regression on mel summary statistics achieves macro-F1 $= 0.311$ (after Bayesian hyperparameter search) with 1,028 parameters, outperforming the best canonical attention transformer (0.246) while using far fewer parameters.
- The performance gap is not closed by regularisation, data augmentation, or larger architectures.
- A cross-linguistic confound explains the val-to-test gap: the validation and test children speak different languages with different class distributions.
- Speaker identity is the dominant variance source, dwarfing random-seed variance.
- Bayesian hyperparameter search (Optuna TPE) over a continuous search space yields three genuine gains over canonical: LR mel ($0.277 \rightarrow 0.311$), LR wav2vec2 ($0.263 \rightarrow 0.285$), and Model 2 wav2vec2 ($0.219 \rightarrow 0.251$). Most transformer searches, especially on mel, confirm that the bottleneck is the data ceiling, not hyperparameter choice.

Dataset and Evaluation Protocol

Dataset

The dataset consists of 1,009 short audio clips extracted from the ACQDIV corpus, annotated with one of four context categories:

Label	Full name	Count
INTER	Social interaction	325 (32.2%)
PLTOY	Playing with toys	240 (23.8%)
EXTEV	External event	222 (22.0%)
MOVE	Movement activity	222 (22.0%)
Total		1,009

Clips come from 9 children spanning three languages: Shipibo-Conibo (South America), Tagalog (Philippines), and Vietnamese (Southeast Asia).

Evaluation Protocol

With 9 participants, the problem setting specifies a **participant-disjoint fixed holdout**:

Split	Children	Clips	Language
Train	7	569	Shipibo, Tagalog, Vietnamese
Validation	1 (child_2)	171	Shipibo
Test	1 (child_6)	269	Tagalog

The validation and test children speak *different languages* — this cross-linguistic mismatch is the central confound discussed in Section 6. The primary metric is **macro-F1**. Final test scores are reported as mean $\pm$ std across 5 random seeds (42, 123, 456, 789, 2024). Seeds are used for variance estimation only, not model selection. Splits are fixed once and never rebuilt.

Methods

Complete Pipeline Overview

Figure 1 summarises the full pipeline. Importantly, **all three feature sources (MFCC, mel, wav2vec2) are evaluated with all three models**. The key distinction is the downstream representation: Model 1 consumes fixed-length summary statistics, whereas Models 2–3 consume padded frame sequences; metadata is used only by Model 3.

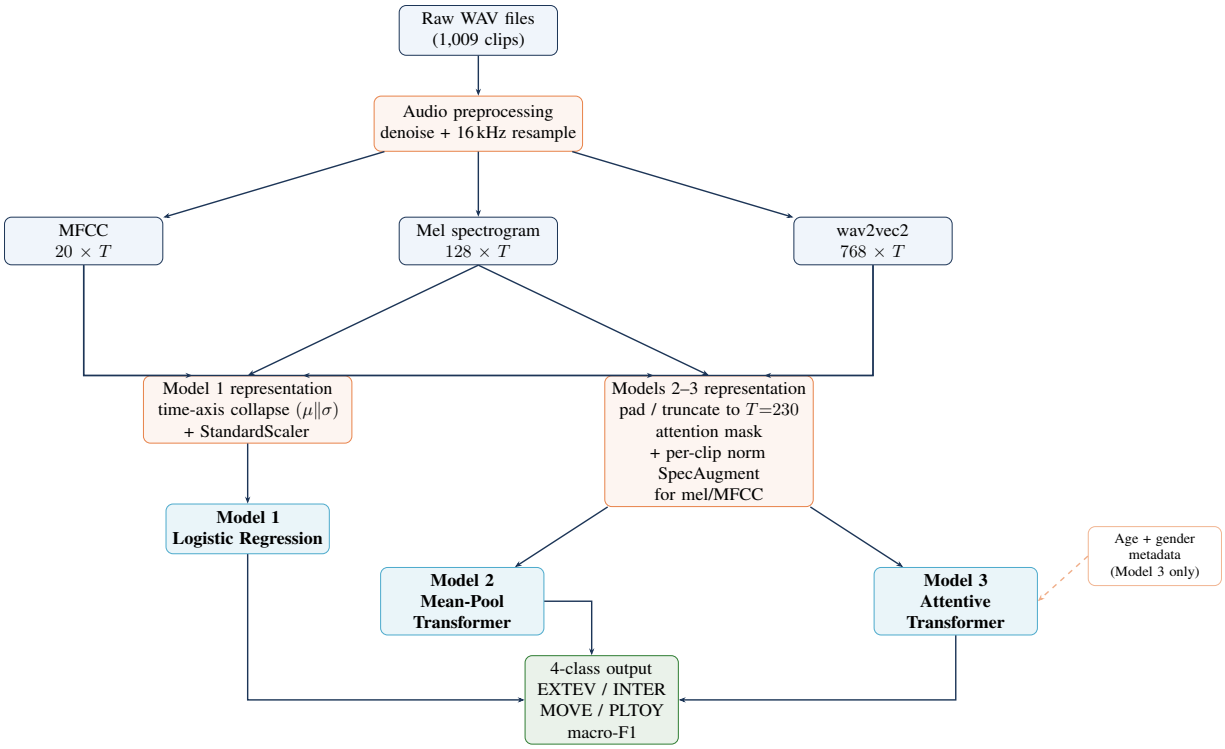


Figure 1: Complete pipeline from raw audio to classification output. All three feature sources (MFCC, mel spectrograms, wav2vec2 embeddings) are evaluated with all three models. Model 1 uses collapsed summary statistics; Models 2–3 use padded sequences, and only Model 3 additionally receives metadata.

Audio-Level Preprocessing

All models share the same preprocessed audio. Raw recordings are first denoised using a spectral noise-reduction procedure applied once and cached. At feature extraction time, each WAV is loaded at 16 kHz using librosa.

Feature Sources

MFCC. 20 Mel-frequency cepstral coefficients extracted via librosa (`n_fft`=2048, `hop_length`=512, 16 kHz). For Model 1, collapsed to a 40-dimensional vector (mean and std per coefficient). For Models 2–3, the padded 128×230 sequence is used (zero-padded to `F_DIM`=128).

Mel spectrogram. 128-bin log-mel spectrogram (`n_fft`=2048, `hop_length`=512, `f_min`=0 Hz, `f_max`=8 kHz, 16 kHz), cached as PNG after denoising. For Model 1, collapses to a 256-dimensional vector (mean and std per bin). For Models 2–3, the padded 128×230 sequence is used.

wav2vec2. Last-layer embeddings from a frozen, pretrained `wav2vec2-base` encoder (768 dimensions per frame).

Model 1: Logistic Regression

The logistic regression baseline receives a fixed-length feature vector and applies: (1) time-axis collapse (mean $\pm$ std), (2) `StandardScaler`, (3) `class_weight='balanced'`, and (4) L2 regularisation strength C

tuned via inner Leave-One-Participant-Out cross-validation. Total parameters: 1,028 (mel), 164 (MFCC), or 6,148 (wav2vec2).

Model 2: Mean-Pool Transformer (Neural Baseline)

An audio-only Transformer encoder followed by masked mean pooling. Each input sequence is per-clip normalised (z-score: subtract per-clip mean and divide by per-clip std across all time–frequency bins) before being fed to the model. SpecAugment [2] is applied during training to spectral sequence inputs (mel spectrograms and MFCCs), with two feature-axis masks ($F_{\max}$=27 bins) and two time masks ($T_{\max}$=70 frames) applied independently per clip. It is not applied to wav2vec2 embeddings. Three independent dropout locations in v2: `dropout_encoder` (inside layers), `dropout_input` (after input projection), `dropout_head` (after pooling).

Model 3: Attentive Transformer (Main Model)

Model 3 replaces mean pooling with metadata-query attentive pooling. The metadata vector (standardised age + one-hot gender) is projected to a query vector $\mathbf{q} \in \mathbb{R}^d$. Attention weights:

$$\boldsymbol{\alpha} = \text{softmax}\left(\frac{\mathbf{q}^\top \mathbf{H}}{\sqrt{d} \cdot \tau}\right), \qquad \hat{\mathbf{h}} = \mathbf{H}\boldsymbol{\alpha}^\top$$

In v2 a fourth dropout location `dropout_meta` is applied to the metadata query after projection.

Training Details

Both transformer models (Models 2 and 3) are trained with the **Adam** optimiser and **cross-entropy loss** with **class weights** set inversely proportional to class frequency (`class_weight='balanced'`), matching the LR baseline. Batch size is **32**. Training runs for up to **100 epochs** with **early stopping** (patience = 15 on validation macro-F1); the best checkpoint by validation F1 is restored before test evaluation. The learning rate is fixed per experiment (no schedule); canonical values are 10^{-3} for mel/MFCC and 5×10^{-4} for wav2vec2. Model 1 (LR) uses the scikit-learn `lbfgs` solver, which converges to a deterministic solution with no stochastic elements.

Detailed Architecture Diagrams

Figures 2–4 show the complete computational graph of each model with exact tensor shapes at each stage. **Pink** nodes are new in v2: up to four independent dropout locations that replace the original single `dropout` parameter (three for Model 2, four for Model 3).

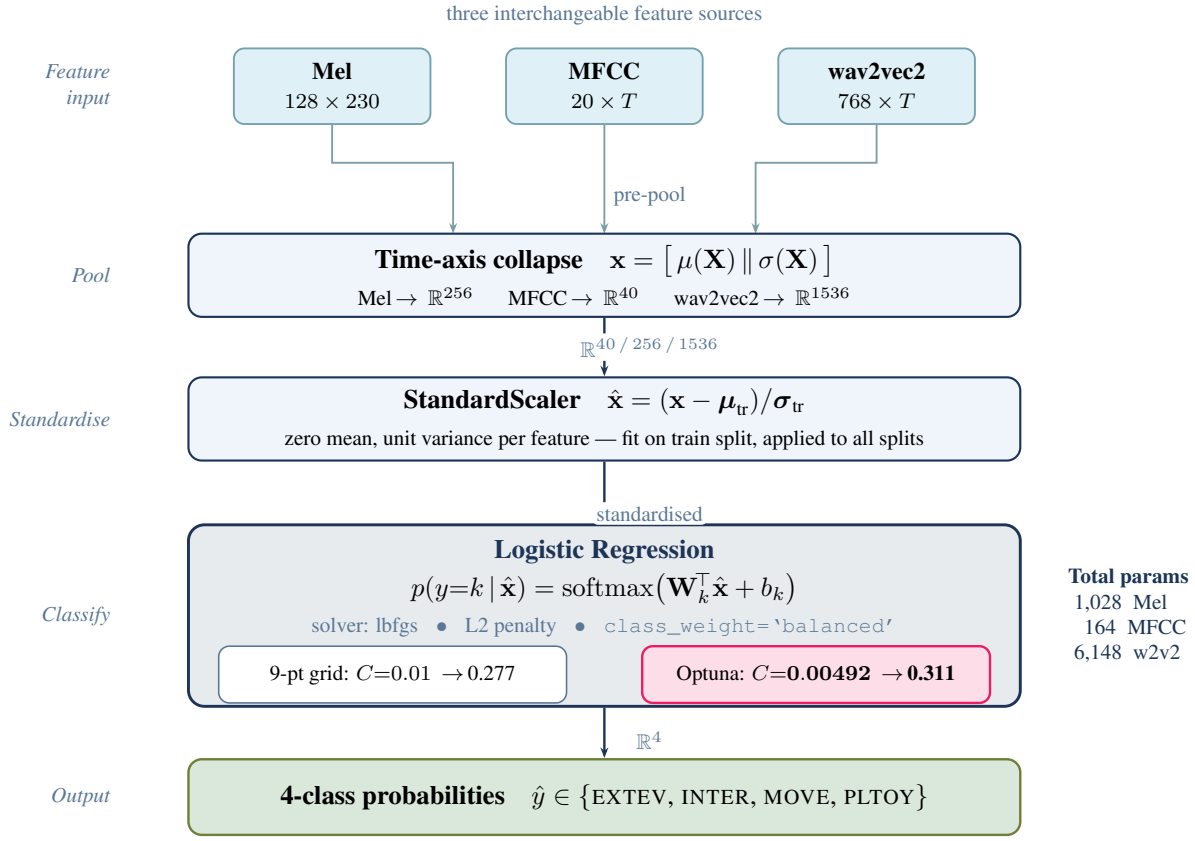


Figure 2: Model 1 (Logistic Regression) architecture. Three feature sources are first collapsed along the time axis via mean ‖ std concatenation, then standardised and passed to a class-balanced L2-regularised classifier. The model is deterministic (σ=0); per-seed variance is zero. The Optuna continuous C line search (pink pill) identified C=0.00492 between discrete grid values, improving mel test macro-F1 from 0.277 to **0.311** — the best result of the entire project.

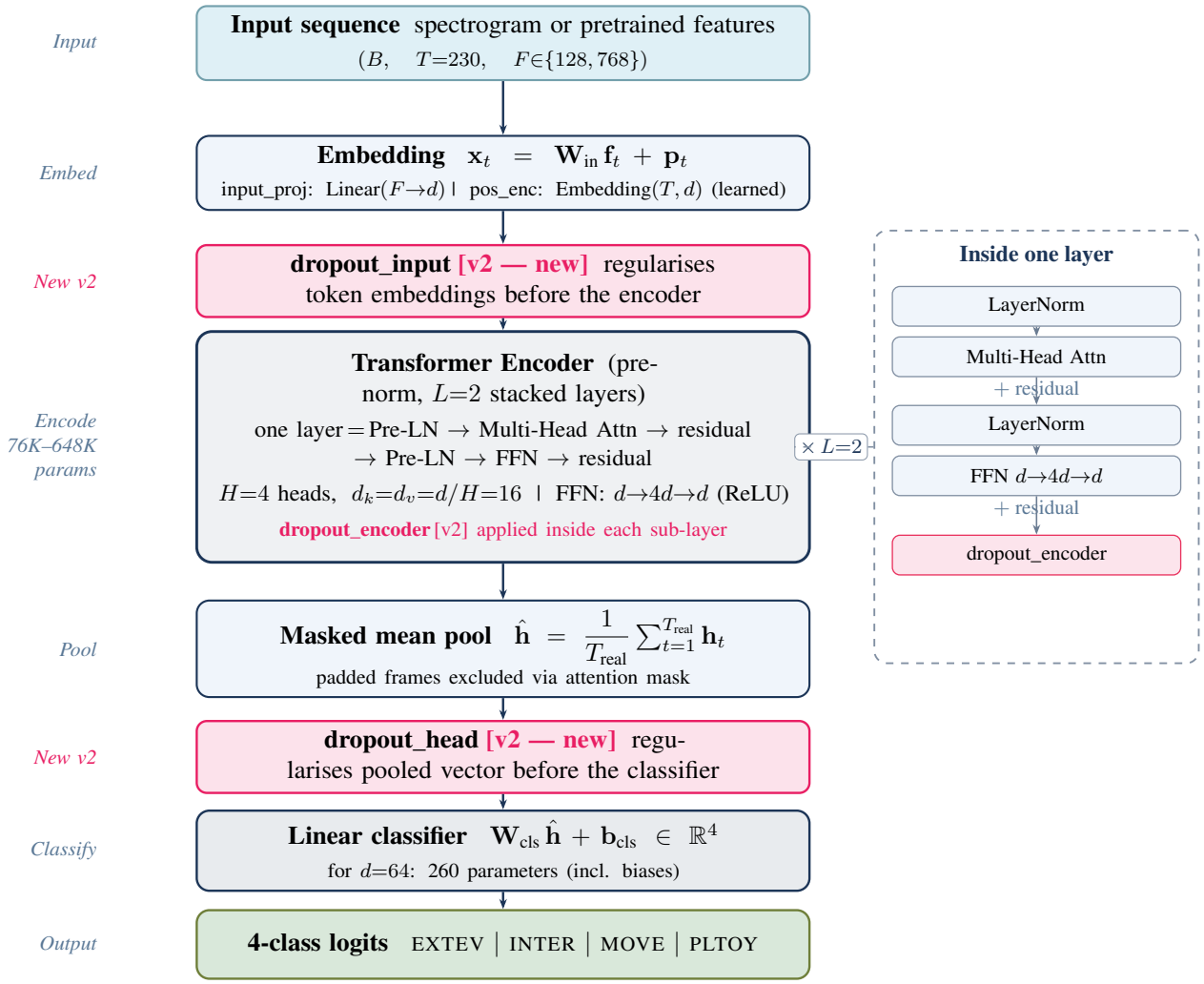


Figure 3: Model 2 (AUDIOTRANSFORMER, mean pooling) v2 architecture. The main computational graph (centre) is annotated with tensor shapes at each stage; the dashed inset (right) reveals the internal sub-layers of one Transformer encoder block, stacked L=2 times. Pink boxes are the three new v2 dropout locations (`dropout_input`, `dropout_encoder`, `dropout_head`) that replace the original single `dropout` parameter. Canonical mel config: d=64, L=2, `dropout_encoder`=0.5, lr=10^{-3} (test F1: 0.249 ± 0.020).

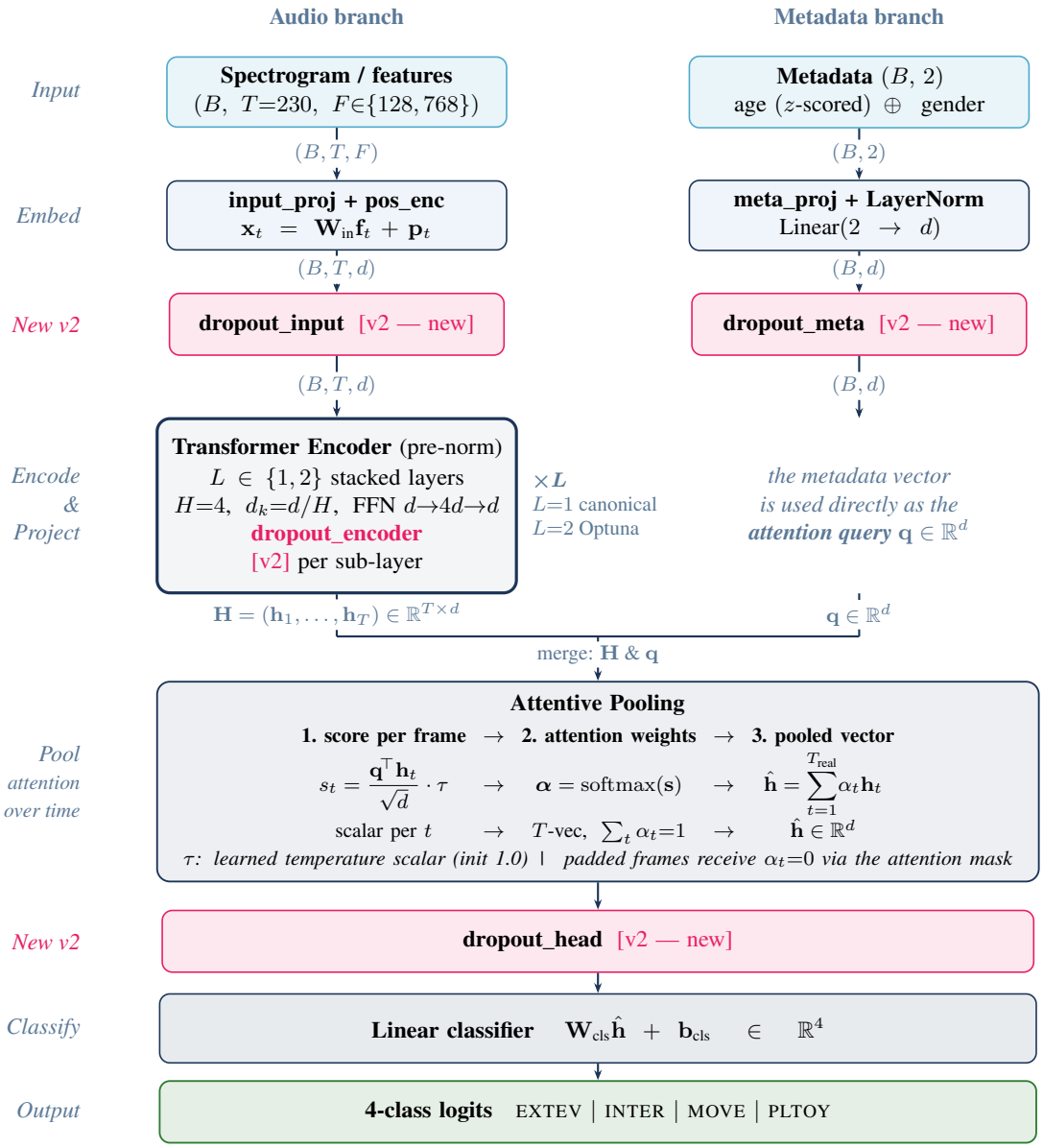


Figure 4: Model 3 (AttentivePoolingTransformer) v2 architecture. The audio branch (left) produces a sequence of contextualised hidden states $\mathbf{H}$; the metadata branch (right) produces a single vector $\mathbf{q}$ that serves as the attention query. Their merge happens inside the **Attentive Pooling** block, which scores every frame against the metadata query, softmaxes into attention weights $\boldsymbol{\alpha}$, and returns the weighted sum $\hat{\mathbf{h}}$. Pink boxes mark the four new v2 dropout locations (`dropout_input`, `dropout_encoder`, `dropout_meta`, `dropout_head`); `dropout_meta` is unique to Model 3. Canonical mel config: d=32, L=1, `dropout_encoder`=0.3 (test F1 0.221 ± 0.013). Optuna config: d=64, L=2, lr=1.24×10^{-4} (test F1 0.233 ± 0.014).

Hyperparameter Tuning

Model 1: Regularisation Strength — Grid Search

The L2 regularisation strength C is selected via inner LOPO cross-validation over the 7 training children. The 9-point logspace grid:

$$C \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}, 10^{4}\}$$

Feature	Selected C	Inner-CV F1	Test F1
MFCC	10	0.243	0.253
Mel	0.01	0.210	0.277
wav2vec2	0.01	0.230	0.263

Transformer Models: Architecture Search

An **8-configuration grid search** per (model, feature source) tunes d_{model}, `num_layers`, `dropout`, and `lr` over the following values:

$d \in \{32, 64\}, \quad L \in \{1, 2\}, \quad$ `dropout` $\in \{0.3, 0.5\}$ (lr fixed at 10^{-3} for mel/MFCC, 5×10^{-4} for wav2vec2)

giving $2 \times 2 \times 2 = 8$ configurations. For Models 2 and 3 on **MFCC and wav2vec2**, the mel-optimised config is reused directly (d=64, L=2 for M2; d=32, L=1 for M3) rather than running a separate search, as the feature dimensionality differences are handled by the shared `input_proj` layer. The wav2vec2 config for M3 was manually adjusted to d=192 after the numerical top-1 showed artefacts from a lucky initialisation (see footnote below).

The three best-performing configurations reported in the table below are the training configurations used for all canonical results:

Model	Feature	d	L	drop	lr	Val F1
Model 2	Mel	64	2	0.5	10^{-3}	0.306
Model 3	Mel	32	1	0.3	10^{-3}	0.286
Model 3	wav2vec2	192	1	0.3	5×10^{-4}	0.297†

† Manually selected; numerical top-1 showed lucky initialisation.

Extension: Bayesian Hyperparameter Search (Optuna TPE)

As an extension we replace the discrete grids with Bayesian optimisation (Optuna Tree-structured Parzen Estimator) across all 9 (model, feature) combinations: 80-trial line searches for LR on each of MFCC, mel, and wav2vec2; 60-trial multi-dimensional searches for M2 and M3 on each of MFCC, mel, and wav2vec2. The transformer searches additionally exercise four independent dropout locations introduced in v2 of the architecture: `dropout_encoder`, `dropout_input`, `dropout_head`, and `dropout_meta` (Model 3 only).

Model 1 — Continuous C Line Search (80 trials per feature)

The discrete grid is replaced by a log-uniform search over $C \in [10^{-6}, 10^{6}]$, seeded with the 9 canonical grid points, and run independently for each of the three feature sources. Optuna identifies $C = 0.00492$ for mel (between the grid values 0.001 and 0.01), $C = 0.01498$ for wav2vec2, and $C = 0.01735$ for MFCC.

Feature	9-pt grid Test F1	Optuna Test F1	Δ	Verdict
Mel	0.277	**0.311**	+0.034	**Genuine gain**
wav2vec2	0.263	**0.285**	+0.022	**Genuine gain**
MFCC	0.253	0.239	−0.014	Genuine loss

Since LR is deterministic (lbfgs solver), no seed averaging is required. Mel yields the largest single gain in the project (+0.034) and establishes **0.311** as the new best result. wav2vec2 also benefits from the finer search (+0.022). MFCC shows the opposite pattern: the inner-CV F1 maximum (0.246 at C=0.01735, beating the grid winner 0.243 at C=10) did not transfer to test performance — a small-scale instance of the "val F1 as unreliable selector" phenomenon documented for the transformers below.

Model 2 — Architecture + Multi-Dropout Search (60 trials per feature)

The search explores d_{model}, L, lr log-uniformly over $[10^{-4}, 10^{-2}]$, and three continuous dropout parameters. For the mel run (April 30 original script), $d \in \{32, 64, 128, 192\}$ and $L \in \{1, 2, 3\}$ were used; for MFCC and

wav2vec2 (rewritten script), $d \in \{32, 64, 128\}$ and $L \in [1, 2]$ were used (192 and L=3 removed as too slow / unstable). The Optuna-selected configurations are reported from 5-seed reruns.

Feature	Best (d, L, lr)	Best trial val/test	Reported test	Canonical	Δ	Verdict
wav2vec2	(128, 1, 8.0×10^{-4})	0.341 / 0.260	**0.251 ± 0.020**	0.219 ± 0.018	$+0.032$	**Genuine gain**
MFCC	(128, 1, 5.1×10^{-3})	0.394 / 0.216	0.219 ± 0.029	0.222 ± 0.033	-0.003	Within seed noise
Mel	(32, 2, 1.83×10^{-3})	0.336 / 0.207	0.226 ± 0.018	0.249 ± 0.020	-0.023	Within seed noise

wav2vec2 (genuine gain). Optuna finds a wider, single-layer architecture with low input/head dropout that improves test F1 by $+0.032$ at $|\Delta|/\sigma = 1.6$. This is the only transformer configuration in the project to genuinely beat its canonical baseline. The canonical val-test gap on wav2vec2 was already small (0.057), leaving room for tuning to add signal rather than overfit.

Mel (within-noise degradation). The single best Optuna trial had val/test $0.336/0.207$, but the corrected 5-seed rerun gives 0.226 ± 0.018 on test. Compared with the canonical 0.249 ± 0.020, this is a drop of $\Delta = -0.023$ and $|\Delta|/\sigma \approx 1.2$, so it is a downward trend but not beyond the seed-noise threshold. Optuna does not improve M2 mel; the canonical M2 mel result (0.249) stands.

MFCC (within noise). The canonical M2 MFCC run (0.222 ± 0.033) confirms the Optuna search adds no signal: the Optuna winner lands at 0.219 ± 0.029, a delta of -0.003 at $|\Delta|/\sigma < 0.1$. The val-test gap of the best Optuna trial (0.394 / 0.216) is by far the largest of any M2 run, indicating heavy overfitting to the Shipibo validation child on this feature source.

Model 3 — Architecture + Multi-Dropout Search (60 trials per feature)

The search adds `dropout_meta` on top of M2's space and uses `MetadataSpectrogramDataset` with proper metadata standardisation.

Feature	Best (d, L, lr)	Best trial val/test	Reported test	Canonical	Δ	Verdict
Mel	(64, 2, 1.24×10^{-4})	0.322 / 0.242	0.233 ± 0.014	0.221 ± 0.013	$+0.012$	Within seed noise
wav2vec2	(64, 1, 1.69×10^{-3})	0.332 / 0.259	0.241 ± 0.021	0.246 ± 0.016	-0.005	Within seed noise
MFCC	(32, 2, 5.2×10^{-4})	0.324 / 0.211	0.217 ± 0.038	0.208 ± 0.040	$+0.009$	Within seed noise

All three M3 Optuna 5-seed results sit within seed noise of their canonical counterparts ($|\Delta|/\sigma < 0.3$ in all cases). Per-seed values for the mel run:

Seed	42	123	456	789	2024	Mean	Std
Val F1	0.263	0.263	0.241	0.232	0.243	0.248	—
Test F1	0.246	0.246	0.239	0.210	0.222	**0.233**	0.014

The marginal mel improvement ($+0.012$) is within the seed-variance noise of $\sigma = 0.013$ but notably reduces the val-test gap from $\sim$0.065 to $\sim$0.016, confirming the lower lr reduces overfitting to the validation child.

Interpretation: Three Genuine Gains, and Val F1 as an Unreliable Selector

Across nine search runs, Optuna achieves three genuine gains: LR mel ($+0.034$), LR wav2vec2 ($+0.022$), and M2 wav2vec2 ($+0.032$). The remaining six are within seed noise of canonical or degrade. The pattern shows that Optuna helps when the validation objective transfers to the test child: LR is deterministic and benefits from finer continuous C values, while M2 wav2vec2 contains enough transferable signal for architecture search to help. When validation–test mismatch is large, especially for M2 mel, Optuna fails to improve transfer and can trend downward on the Tagalog test child.

The transformer-on-mel evidence reinforces this directly: across 60 trials each for M2 and M3, higher validation F1 does not predict higher test F1 (Figure 8). The Shipibo validation child and the Tagalog test child have different class distributions, so a model tuned to fit the Shipibo prior generalises worse to Tagalog

— a finding that is structural and cannot be solved by hyperparameter search alone (see Section 6).

Experiments and Results

Main Results

Table 1: Test macro-F1 for all model–feature combinations. Multi-seed rows are shown as mean ± std over 5 seeds; LR is deterministic (std = 0.000). Optuna gains over canonical are shown in **pink**; genuine losses in gray; within-noise changes are unbolded.

Model	MFCC	Mel	wav2vec2
Model 1 (LR) canonical (9-pt grid)	**0.253** ± 0.000	0.277 ± 0.000	0.263 ± 0.000
Model 1 (LR) + Optuna	0.239 ± 0.000	**0.311** ± 0.000	**0.285** ± 0.000
Model 2 (mean pool) canonical	0.222 ± 0.033	0.249 ± 0.020	0.219 ± 0.018
Model 2 (mean pool) + Optuna	0.219 ± 0.029	0.226 ± 0.018	**0.251** ± 0.020
Model 3 (attention) canonical	0.208 ± 0.040	0.221 ± 0.013	**0.246** ± 0.016
Model 3 (attention) + Optuna	0.217 ± 0.038	0.233 ± 0.014	0.241 ± 0.021
Chance (uniform)		0.245 ± 0.027	

Key findings:

- The Optuna line search for LR mel yields $\mathbf{0.311}$, the best result of the project — a 1,028-parameter linear model outperforms a 126K-parameter mean-pool transformer by 24% and a 648K-parameter attention transformer by 26%.
- Model 2 wav2vec2 + Optuna (0.251 ± 0.020) is the **only transformer configuration to genuinely improve over its canonical baseline** ($+0.032$ at $|\Delta|/\sigma = 1.6$), and is one of the three genuine Optuna gains in the project.
- LR + Optuna improves on every feature where the canonical gap was already small (mel, wav2vec2) and degrades on MFCC, where the inner-CV maximum did not transfer to test.
- M3 mel and M3 wav2vec2 + Optuna are within seed noise of canonical; M2 mel + Optuna trends downward by 0.023 but remains within seed noise; the corrected 5-seed val–test gap is about 0.065.

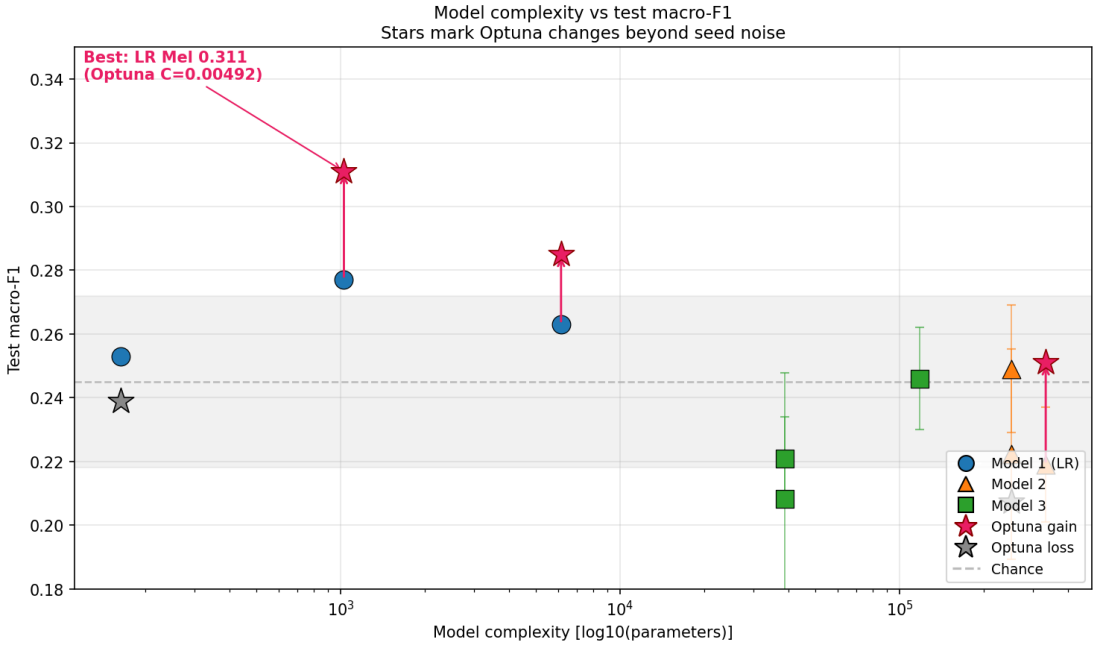


Figure 5: Test macro-F1 vs. model complexity (log scale). Stars ($\star$) show Optuna changes beyond the noise threshold; pink stars mark genuine gains and gray stars mark genuine losses. LR mel reaches $\mathbf{0.311}$ (pink star, top); LR wav2vec2 reaches 0.285; and M2 wav2vec2 + Optuna reaches 0.251 (pink star, mid-right). More parameters do not improve accuracy.

Per-class Breakdown

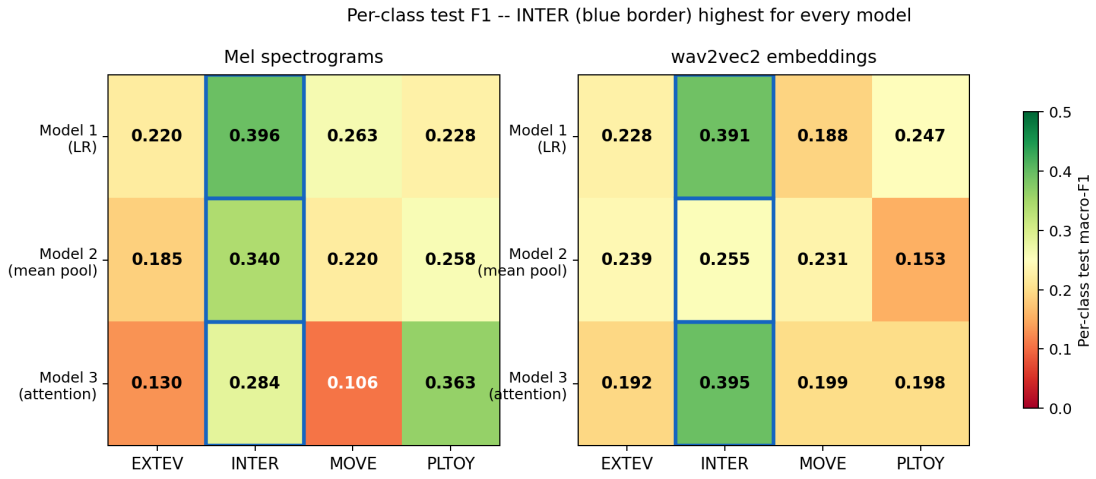


Figure 6: Per-class test F1 heatmap. INTER (blue border) is the highest-F1 class in five of six panels; the exception is Model 3 on mel, where PLTOY is highest.

The pattern is strikingly uniform across architectures: INTER is the highest-F1 class in five of six panels, with Model 3 on mel as the exception (PLTOY is highest there). EXTEV and MOVE are generally lower and unstable across models. The similarity between a 1,028-parameter LR and larger transformer baselines is strong evidence that the ceiling is *data-driven*.

The INTER Over-Prediction Problem

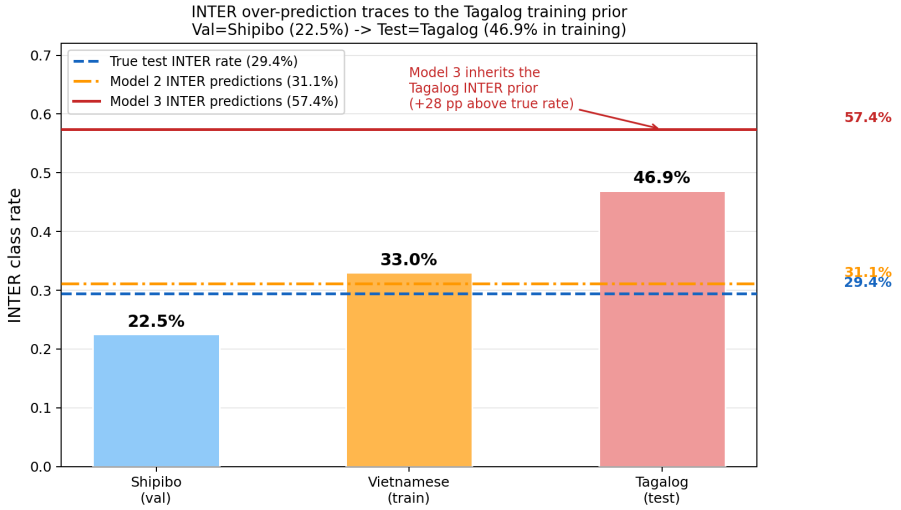


Figure 7: INTER class rate by language in training data (bars) and model prediction rates (horizontal lines). Model 3 predicts INTER on 57.4% of test clips vs. the true rate of 29.4%.

Two compounding effects drive the INTER bias: (1) the Tagalog training prior (46.9% INTER) is substantially higher than the true test rate (29.4%); and (2) Model 3 on wav2vec2 attends to a single frame in 95% of clips, which most likely carries interaction-like content.

Extension: Optuna Search Outcomes

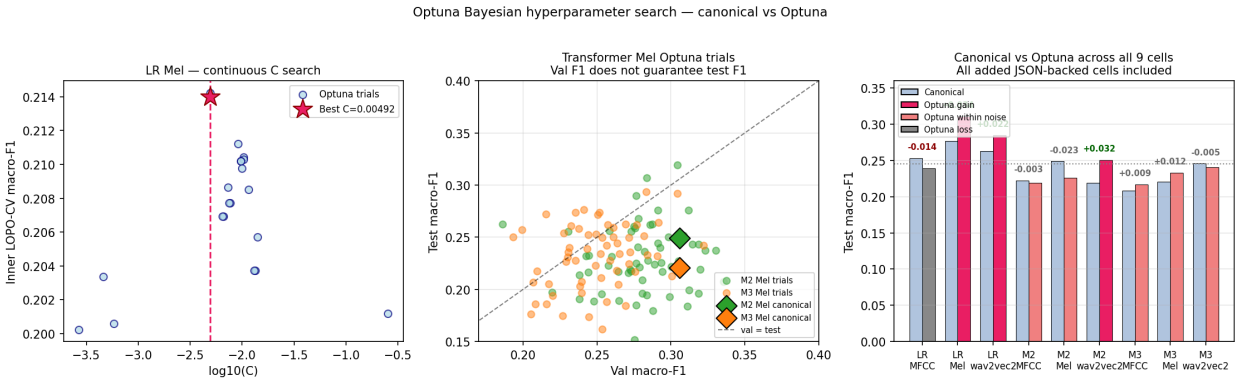


Figure 8: **Left**: LR mel continuous C search — best C=0.00492 lies between discrete grid points. **Centre**: M2/M3 mel Optuna trials — val F1 does not predict test F1 (language confound). **Right**: canonical vs. Optuna for all 9 (model, feature) cells. Three genuine gains appear in pink: LR mel (+0.034), LR wav2vec2 (+0.022), and M2 wav2vec2 (+0.032).

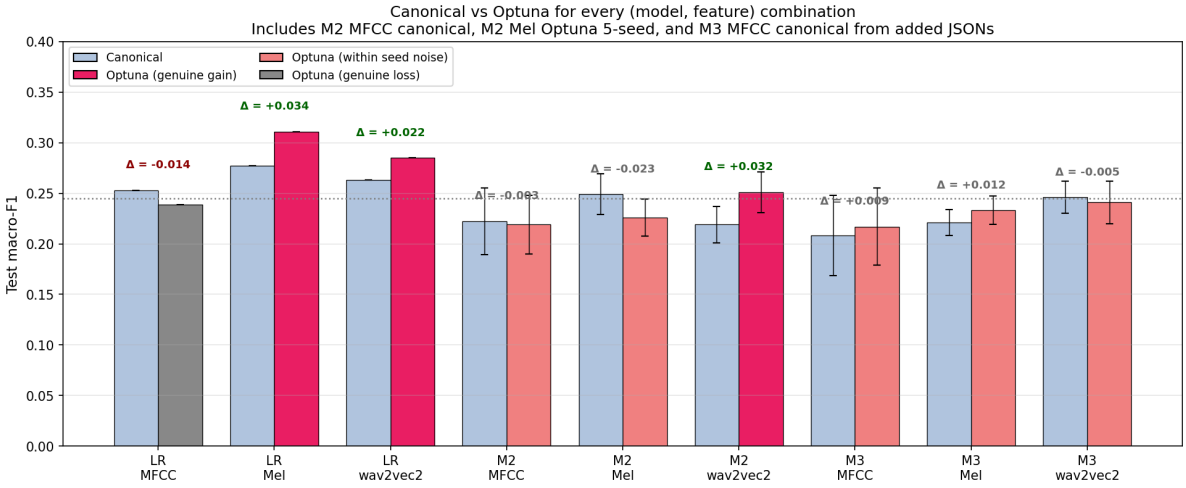


Figure 9: Canonical vs. Optuna for every (model, feature) cell, with Δ annotated above each pair. Three cells show genuine gains: LR mel (+0.034, deterministic), LR wav2vec2 (+0.022, deterministic), and M2 wav2vec2 (+0.032 at $|\Delta|/\sigma$=1.6). LR MFCC is the only genuine loss; M2 mel trends downward but remains within seed noise, as do the remaining non-gain transformer cells.

The 9-cell view in Figure 9 consolidates the extension's findings. Optuna's success splits along a clear axis: it adds signal where the validation signal transfers to the test child, and it overfits the validation child where the split mismatch is large. For deterministic LR, the mel and wav2vec2 improvements are not seed artifacts; for transformers, only M2 wav2vec2 genuinely improves because its canonical configuration was already well calibrated. M2 mel trends downward within seed noise, while the M3 mel/wav2vec2 configurations show within-noise changes, consistent with the language-confound interpretation in Section 6.

Architecture Comparison: Models 2 vs. 3

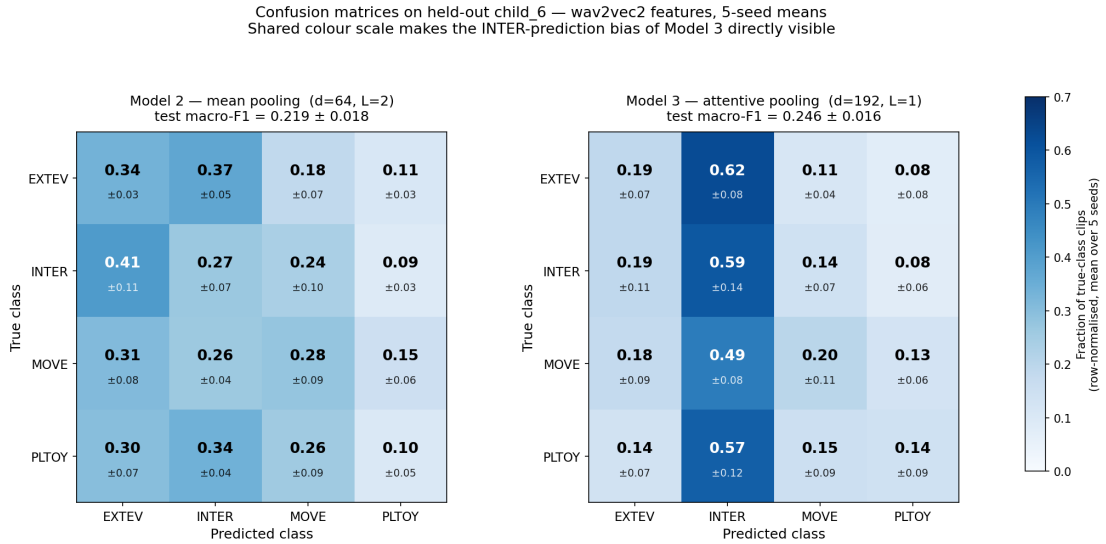


Figure 10: Row-normalised confusion matrices for Model 2 (left) and Model 3 (right) on wav2vec2, 5-seed means. Model 3 predicts INTER 57.4% of the time vs. the true 29.4%.

Error Analysis: Data-Driven Ceiling

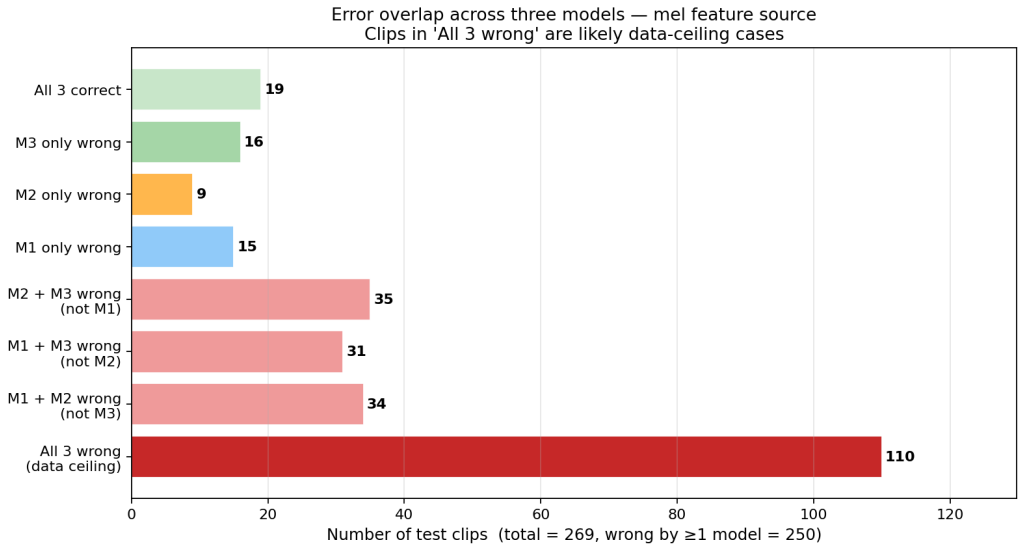


Figure 11: Error overlap: 110 clips (40.9%) are wrong by all three models simultaneously; only 19 (7.1%) are correct by all three.

The three models have nearly identical per-clip error rates (M1: 70.6%, M2: 69.9%, M3: 71.4%) and the same 110 clips fail all three architectures.

Regularisation Sweep

A 6-configuration sweep on mel features varying dropout (0.3–0.7), weight decay (10^{-4}, 5×10^{-4}), and label smoothing (0.0–0.2) achieved 0.240 ± 0.028, within seed noise of the canonical 0.221. The Optuna extension corroborates this on mel features: 60 Bayesian-guided trials for both M2 and M3 yield no improvement beyond seed noise. On mel, the data ceiling dominates. On wav2vec2, however, M2 + Optuna does break out of canonical performance ($0.219 \rightarrow 0.251$), suggesting the ceiling is feature-source-specific rather than universal.

Discussion

Why Does LR Beat Transformers?

Mel summary statistics capture the average spectral content of a clip — a surprisingly strong signal for context classification. With only 569 training clips from 7 children, the transformer learns speaker identity as effectively as context identity. Mean pooling over the whole clip effectively ignores temporal dynamics, making Model 2 a noisier version of the LR signal.

When Did Optuna Improve a Model, and When Didn't It?

The discrete 9-point grid for LR ($C \in [10^{-4}, 10^{4}]$, logspaced) was too coarse in the region $[0.001, 0.01]$. The optimal $C = 0.00492$ for mel and $C = 0.01498$ for wav2vec2 lie between grid points — continuous search found them, discrete search could not. These are genuine signal-to-noise ratio improvements.

More surprisingly, **Model 2 on wav2vec2** also benefited from Optuna ($+0.032$ at $|\Delta|/\sigma = 1.6$). This is the only transformer configuration in the project to genuinely beat its canonical baseline. The canonical val-test gap for M2 wav2vec2 was small (comparable to LR), leaving room for the search to add signal rather than overfit the validation child.

For the remaining transformer searches, Optuna found configurations with higher validation F1 but lower or unchanged test F1. The clearest example is M2 mel: the 5-seed rerun gives 0.226 ± 0.018, which is 0.023 below the canonical 0.249 ± 0.020 but still within seed noise. The corrected val–test gap is about 0.065, showing that the search did not transfer reliably from the Shipibo validation child to the Tagalog test child.

This is not simply a hyperparameter problem — it is the language confound. LR MFCC shows a smaller-scale version of the same pattern: inner-CV F1 maximum at C=0.01735 (0.246) beat the grid winner (0.243) but transferred to a slightly lower test F1 (0.239 vs. 0.253).

The pattern is consistent across the nine searches: Optuna helps when the validation signal transfers to test, and otherwise gives within-noise changes or degradation.

The Language Confound

The validation child (child_2) speaks Shipibo-Conibo (22.5% INTER in training). The test child (child_6) speaks Tagalog (46.9% INTER in training). Models tuned on a Shipibo validation set encounter a distributional shift at test time. The Optuna results provide direct evidence: across 60 transformer trials, higher validation F1 does not predict higher test F1 (Figure 8, centre panel). This confirms our interpretation — the confound is structural and cannot be solved by hyperparameter search.

Speaker Identity is the Dominant Bottleneck

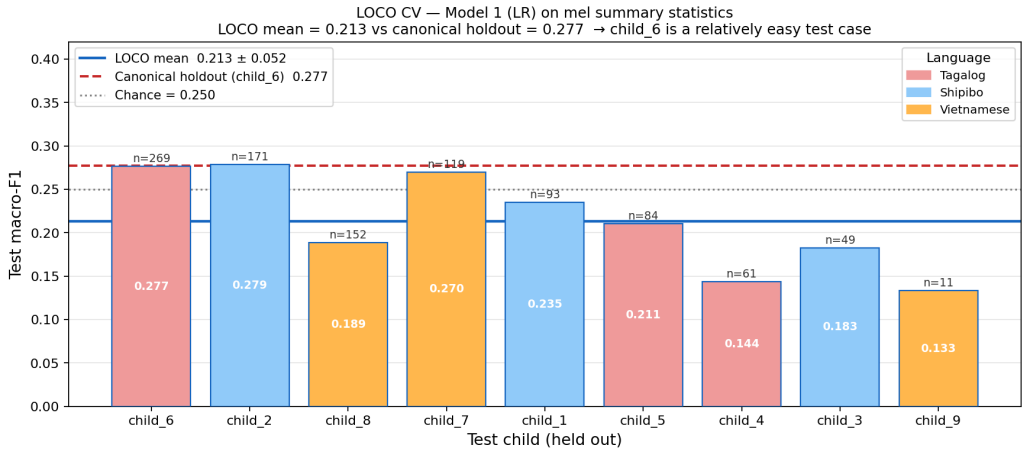


Figure 12: LOCO cross-validation for Model 1 (LR) on mel features. The canonical holdout (child_6) is a relatively easy test case. LOCO mean = 0.213 ± 0.053.

The LOCO mean F1 is 0.213 ± 0.053, below the fixed child_6 holdout score (0.277 with the canonical grid and 0.311 after Optuna). Per-fold range: 0.133 (child_9) to 0.279 (child_2). Speaker variance ($\sigma \approx 0.053$) dominates LR seed variance, which is zero because the lbfgs solver is deterministic.

wav2vec2 Fine-Tuning Does Not Help

A controlled experiment compared frozen wav2vec2 feature extraction against partial fine-tuning of the top 2 and top 4 transformer layers (1 seed, 35 epochs, early stopping patience = 8). The frozen baseline reproduces the canonical M2 wav2vec2 result closely (test F1 = 0.218 vs. canonical 0.219 ± 0.018), confirming the experimental setup is consistent with the main results.

Training F1 rose monotonically as layers were unfrozen: from 0.462 (frozen) to 0.636 (top-2) and 0.542 (top-4). This is the signature of speaker memorisation: with only 7 training children, additional trainable parameters provide more capacity for learning who each child sounds like, not more capacity for learning what context they are in.

The val-test gap is the clearest diagnostic. For the frozen baseline the gap is negligible (+0.006). Unfreezing the top-2 layers widens it to +0.059 — a factor of 10× — and unfreezing the top-4 layers widens it further to +0.077 (13×). The fine-tuned model learns to fit the Shipibo validation child more closely, but that knowledge does not transfer to the Tagalog test child. This is the language confound expressed directly in the training dynamics.

The apparent +0.040 test-F1 improvement for top-2 (0.258 vs. frozen 0.218) is not reliable. Test F1 oscillated

between 0.124 and 0.307 across epochs while training F1 climbed monotonically to 0.730; early stopping selected epoch 28 by Shipibo validation F1 (0.317), biasing the checkpoint toward the validation language. Even at this selected checkpoint, fine-tuning with 14 million additional trainable parameters did not reach the performance of simple logistic regression on frozen wav2vec2 features (0.263, canonical).

These results confirm the prediction from the LOCO analysis: with $\sigma_{\text{speaker}} \approx 0.053$ dominating all other variance sources, increasing the number of trainable parameters amplifies speaker-identity overfitting rather than improving context classification.

What Attention Concentration Tells Us

On wav2vec2, Model 3's attention collapses to a single frame in 95% of clips — the model uses one contextualised embedding per clip rather than the full sequence. On mel, attention spans $\sim$17 frames. The learned temperature $\tau < 1.0$ in both configurations indicates the model tried to soften attention; the concentration on wav2vec2 is driven by score magnitudes, not temperature runaway.

Future Work

Larger and more diverse participant pool. The LOCO analysis shows the bottleneck is speaker diversity, not model capacity.

Matched-language validation strategy. The language confound suggests future work should control for language distribution when assigning children to train/val/test splits.

Partial wav2vec2 fine-tuning. A controlled experiment (1 seed, 35 epochs) confirmed that unfreezing the top 2–4 transformer layers amplifies speaker-identity overfitting on this dataset: the val-test gap widened up to $13\times$ while test F1 remained flat or within noise. A child-speech pretrained encoder (e.g. wav2vec2 fine-tuned on CHILDES) used as a frozen extractor would be a more principled next step before attempting fine-tuning at this data scale.

LOCO evaluation for all models. We ran LOCO only for Model 1 due to compute constraints. Running for Models 2 and 3 would reveal whether the language prior affects transformer models more severely.

Richer context features. Contextual metadata such as recording setting (home, outdoor, playgroup) could improve classification without exposing participant identity.

Conclusion

We presented a controlled comparison of three models for classifying child vocalisation contexts across three under-resourced language settings. Logistic regression on mel summary statistics outperforms both transformer architectures — a result robust across feature sources, random seeds, pooling strategies, and LOCO cross-validation. As an extension, Bayesian hyperparameter search (Optuna) yields three genuine gains over canonical: LR mel from 0.277 to **0.311**, LR wav2vec2 from 0.263 to 0.285, and Model 2 wav2vec2 from 0.219 to 0.251. The remaining six (model, feature) searches are within seed noise of canonical or degrade — consistent with the language confound, the 40.9% all-models-wrong rate, and the LOCO-derived speaker-variance estimate. The findings underscore that in small, speaker-limited datasets, model simplicity and careful hyperparameter search outweigh architectural complexity, but that the benefit of careful search depends on whether the validation set is representative of test.

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Supplementary Tables and Figures

Per-seed Results

Table 2: Per-seed test macro-F1 for canonical and Optuna-selected runs. LR std = 0.000 (lbfgs is deterministic). Transformer ⋆ rows are 5-seed reruns of Optuna-selected configurations; the M2–Mel row uses the corrected multiseed JSON.

Model – Feature	s=42	s=123	s=456	s=789	s=2024	Mean±Std
LR – MFCC	0.253	0.253	0.253	0.253	0.253	0.253±0.000
LR – MFCC ⋆	0.239 (deterministic; C=0.01735)					0.239±0.000
LR – Mel	0.277	0.277	0.277	0.277	0.277	0.277±0.000
LR – Mel ⋆	**0.311 (deterministic; C=0.00492)**					**0.311±0.000**
LR – wav2vec2	0.263	0.263	0.263	0.263	0.263	0.263±0.000
LR – wav2vec2 ⋆	**0.285 (deterministic; C=0.01498)**					**0.285±0.000**
M2 – MFCC	0.242	0.216	0.267	0.219	0.168	0.222±0.033
M2 – MFCC ⋆	0.188	0.216	0.235	0.265	0.189	0.219±0.029
M2 – Mel	0.257	0.242	0.230	0.277	0.238	0.249±0.020
M2 – Mel ⋆	0.220	0.248	0.202	0.212	0.247	0.226±0.018
M2 – wav2vec2	0.210	0.237	0.209	0.239	0.200	0.219±0.018
M2 – wav2vec2 ⋆	0.246	0.236	0.279	0.267	0.225	**0.251±0.020**
M3 – MFCC	0.205	0.138	0.237	0.209	0.253	0.208±0.040
M3 – MFCC ⋆	0.244	0.248	0.180	0.163	0.251	0.217±0.038
M3 – Mel	0.228	0.211	0.218	0.231	0.217	0.221±0.013
M3 – Mel ⋆	0.246	0.246	0.239	0.210	0.222	0.233±0.014
M3 – wav2vec2	0.250	0.262	0.241	0.263	0.214	0.246±0.016
M3 – wav2vec2 ⋆	0.234	0.282	0.222	0.229	0.239	0.241±0.021

⋆ Optuna-selected configuration. Bold = genuine gain over canonical.

Optuna Search Parameters

Table 3: Optuna search configuration for each (model, feature) combination. Transformer searches share the same lr/dropout structure. The M2 mel run (April 30, original script) used $d \in \{32, 64, 128, 192\}$ and $L \in \{1, 2, 3\}$; M2 MFCC and wav2vec2 used $d \in \{32, 64, 128\}$ and $L \in [1, 2]$; M3 adds `dropout_meta`.

Search	Trials	Search space	Objective
LR – MFCC	80	C log-uniform $[10^{-6}, 10^{6}]$; seeded with 9-pt grid	Inner LOPO-CV macro-F1
LR – Mel	80	ditto	Inner LOPO-CV macro-F1
LR – wav2vec2	80	ditto	Inner LOPO-CV macro-F1
M2 – MFCC	60	shared transformer space (3 dropouts)	Val macro-F1
M2 – Mel	60	$d \in \{32, 64, 128, 192\}$, $L \in \{1, 2, 3\}$, 3 dropouts	Val macro-F1
M2 – wav2vec2	60	shared transformer space (3 dropouts)	Val macro-F1
M3 – MFCC	60	shared + `dropout_meta`	Val macro-F1
M3 – Mel	60	shared + `dropout_meta`	Val macro-F1
M3 – wav2vec2	60	shared + `dropout_meta`	Val macro-F1

Ablation Summary

Table 4: Summary of all supplementary ablations and their test F1 outcomes. Top section: feature/architecture ablations. Bottom: full Optuna extension results across all 9 (model, feature) combinations.

Ablation	Baseline F1	Change
Feature / architecture / metadata ablations:		
Delta-MFCC features (LR MFCC)	0.253	-0.024
Seed ensembling (M2 Mel)	0.249	-0.024
Metadata: age only (M3 wav2vec2)	0.247	$+0.001$
Metadata: gender only	0.221	-0.025
Metadata: none (no conditioning)	0.233	-0.013
Reg. sweep best (M2 wav2vec2)	0.219	$+0.021$ (val only)
Optuna extension (all 9 cells):		
Optuna LR MFCC (C=0.01735)	0.253	-0.014
Optuna LR Mel (C=0.00492)	0.277	$\mathbf{+0.034}$ **(genuine)**
Optuna LR wav2vec2 (C=0.01498)	0.263	$\mathbf{+0.022}$ **(genuine)**
Optuna M2 MFCC 5-seed	0.222	-0.003 (within noise)
Optuna M2 Mel 5-seed	0.249	-0.023 (within noise, $\lvert\Delta\rvert/\sigma \approx 1.2$)
Optuna M2 wav2vec2 5-seed	0.219	$\mathbf{+0.032}$ **(genuine,** $\lvert\Delta\rvert/\sigma$=1.6**)**
Optuna M3 MFCC 5-seed	0.208	$+0.009$ (within noise)
Optuna M3 Mel 5-seed	0.221	$+0.012$ (within noise)
Optuna M3 wav2vec2 5-seed	0.246	-0.005 (within noise)
wav2vec2 fine-tuning experiment (1 seed, 35 epochs, seed 42):		
Frozen wav2vec2 + head (baseline)	0.218	—
Fine-tune top 2 layers + head	0.218	$+0.040$ (unreliable — val-test gap $\times 10$: $+0.006 \rightarrow +0.059$)
Fine-tune top 4 layers + head	0.218	$+0.004$ (within noise — val-test gap $\times 13$: $+0.006 \rightarrow +0.077$)

LOCO Per-fold Results

Table 5: LOCO cross-validation for Model 1 (LR) on mel features. LOCO mean = 0.213 ± 0.053; canonical grid holdout = 0.277 (Optuna holdout = 0.311).

Child	N clips	Language	Test F1
child_6	269	Tagalog	0.277
child_2	171	Shipibo	0.279
child_7	119	Vietnamese	0.270
child_1	93	Shipibo	0.235
child_5	84	Tagalog	0.211
child_8	152	Vietnamese	0.189
child_3	49	Shipibo	0.183
child_4	61	Tagalog	0.144
child_9	11	Vietnamese	0.133
Mean			**0.213 ± 0.053**